

Evaluating the Risk to Bulk Power System Reliability from Large Load Induced Oscillations

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Abstract—Unexpected operations of large loads (LL) like hyperscale data centers may threaten grid reliability. Computation-cycle-driven load fluctuations may introduce forced oscillations (FOs) that can get amplified due to resonance with natural modes of the power system. Planning studies informing the load interconnection processes currently do not account for these LL-induced oscillations, leaving a gap in the reliability evaluations. To address this, we propose a simulation-based risk assessment framework that quantifies the wide-area grid reliability impact of LL-induced oscillations as a function of FO frequency and source location. The framework is illustrated using a 240-bus transmission network model with 13 potential LL interconnection locations. Multiple reliability metrics are computed, including the extent of spatial propagation of FOs, risks of modal resonance, and the severity of oscillation amplification. The proposed methodology enables system planners to pinpoint high-risk LL interconnection points, FO frequencies of concern, and determine targeted mitigation strategies.

Index Terms—large electric load, data center, forced oscillation, reliability

I. INTRODUCTION

The power grid is experiencing an era of rapid load growth due to the increase in energy demand from data centers, driven by a global race for dominance in artificial intelligence (AI). Forecasts estimate that data center energy usage may grow from $\sim 4.4\%$ of the total US electricity consumption in 2023 to $\sim 12\%$ by 2028 [1]. Grid operators worldwide are grappling with large load (LL) interconnection requests and the unique operational risks posed by the dynamic behavior of these loads.

Hyperscale data centers can consume hundreds of MW at a single load site, with GW-scale plants also being planned [2]. Their power demand can be cyclic, characterized by periodic consumption bursts and cool-down periods. Traditionally, grid planning has not accounted for risks posed by the widespread deployment of such hyperscalers. Thus, unexpected LL behavior can threaten reliability by introducing or exacerbating grid disturbances [3]. The North American Electric Reliability Corporation (NERC) Large Load Task Force recently published a white paper outlining key reliability concerns posed by emerging LLs like AI-training facilities [4]. The paper identifies several high-priority risks, including forced oscillations (FOs) driven by periodic AI workloads.

Power system oscillations may be natural or forced. Natural oscillations occur when system modes are excited by transient disturbances or random load fluctuations, while FOs are driven by periodic external disturbances. They persist in the system as long as the external forcing function is present and

pose multiple reliability risks. High-amplitude oscillations may cause incorrect protection/control actions leading to cascading failures. High-frequency oscillations may cause flicker or interact with turbine torsional modes, increasing generator fatigue. Power swings may be amplified away from the source due to resonance if the FO frequency is close to a system mode [5]. Besides the risk of inadvertent protection actions, resonance further complicates the problem of locating and isolating the FO source.

The threat of oscillatory instability is well recognized and grid operators employ various solutions to monitor and mitigate FOs during operations [5], [6]. It is assumed that these oscillations occur when control equipment malfunction, and are not a feature of normal load operations. Therefore, the system-wide reliability impact of load-induced oscillations is typically not studied during planning. However, given the cyclic nature of large computational loads, assessing the impact of FOs has become critical at the planning stage itself.

At present, there is a lack of established study procedures to evaluate the wide-area reliability impact of load-induced low-frequency oscillations. To address this gap, this paper proposes a simulation-based, readily automatable risk-evaluation methodology that quantifies the impact of FOs as a function of their frequencies and source location. Oscillation impact is quantified by several metrics like the spatial spread of FOs, risk of modal resonance, and the severity of oscillation amplification. Using the proposed framework, high-risk LL interconnection points and critical oscillation frequencies are identified in the reduced 240-bus model of the Western Electricity Coordinating Council (WECC) system [7]. Simulations verify that the degree of oscillation amplification observed is directly correlated with modal observability at the source.

The remaining paper is organized as follows. Section II describes how generic load models can be used to simulate FOs in positive sequence studies. The proposed risk assessment methodology is presented in Section III and used to study the impact of LL-induced oscillations on the 240-bus WECC model in Section IV. Section V discusses findings, outlines future research directions, and concludes this paper.

II. MODELING DATA CENTER BEHAVIOR

The major load components in a typical data center include (a) servers, (b) cooling load for removing heat dissipated in server racks, and (c) auxiliary load like lighting. Data centers are equipped with uninterruptible power supplies (UPS) to protect voltage-sensitive equipment and may also have back-

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TABLE I
LOAD COMPOSITION IN THE CMLD MODEL USED TO REPRESENT DATA CENTER DYNAMICS

CMLD Component	Description	Data Center Component	Typical Parameter Setting
Motor A	3 ϕ motors with constant torque compressors	Cooling Load	$FMA = 0.09$
Motor B	3 ϕ motors with variable torque compressors (fans)	N/A	$FMB = 0$
Motor C	3 ϕ motors with variable torque compressors (pumps)	N/A	$FMC = 0$
Motor D	1 ϕ motor load	N/A	$FMD = 0$
Electronic Load	Power electronic loads	Server load & CRAH	$FEL = 0.89, PFEL = 1$
Static Load	Generic ZIP load	Miscellaneous load	$PFs = 1, P1e = 0, P2e = 0, P1c = 1, P2c = 0$

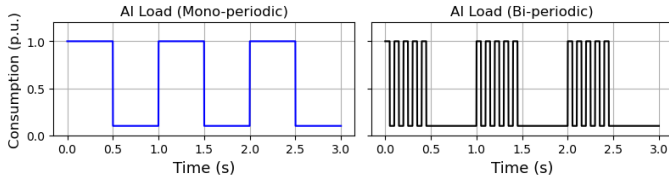


Fig. 1. Cyclic power consumption patterns of large computational loads

up generation. Several high-fidelity device-level data center models have been developed [8], [9], but these are not well-suited for interconnection-scale planning studies where networks with thousands of buses need to be modeled. Current industry practice represents data center power consumption with standard PQ or voltage-dependent loads that do not capture behavior like load ramps and delayed reconnection after grid disturbances. Ongoing modeling efforts are attempting to integrate these data center-specific features, but these models are not yet available in commercial simulation engines [10], [11]. Thus, system-level planning studies must still rely on generic load models like ZIP (constant impedance, current, power) and composite load model (CMLD) [12]. Heeding this constraint, this work utilizes generic load models to emulate the LL cyclic consumption pattern, as elaborated next.

A. Cyclic Consumption Pattern

Data center power consumption can be irregular, dictated by the computational workload. In AI-training facilities, synchronized (de)activation of large arrays of graphical and tensor processing units creates a cyclic demand pattern characterized by quick bursts of consumption followed by cool-down periods. The demand swings may be as high as from 10% to 100% of full load [13]. AI-training load cycles may also be bi-periodic. Slower variations (0.1–1 Hz) reflect rest-to-compute transitions, and faster fluctuations (5–30 Hz) within compute phases occur due to rapid switching of individual modules [13]. These load consumption patterns have been illustrated in Fig. 1. Oscillations can also originate from auxiliary equipment. FOs at cryptocurrency mining facilities caused by communication and firmware failures have been reported [4]. In a data center within Dominion Energy’s footprint, power consumption bursts every second were found to excite a 14.7-Hz mode caused by UPS control settings [14].

B. Simulating FOs with Generic Load Models

1) *ZIP Model*: ZIP models comprise a combination of constant impedance, current, and power components [15]. FOs can be simulated by periodically scaling load values during dynamic simulation. Popular positive sequence simulators offer commands to automate such implementations. For

example, in the Python API for Siemens PTI PSS/E v35, the `load_change_6()` command can be used.

2) *Composite Load Model (CMLD)*: CMLD models consist of several motor, electronic, and static load components. NERC recently conducted an industry-wide survey with developers and vendors to understand load composition within a data center [16]. Based on the findings, the CMLD composition in Table I is recommended for modeling typical data center dynamics. Once the appropriate CMLD parameters are configured, periodic load fluctuations can be simulated using the same approach as the ZIP model.

III. METHODOLOGY

Sustained FOs pose reliability risks ranging from operator confusion to relay misoperations. The risk is higher when FOs propagate to wide areas instead of having a localized impact. The wide-area spread is likely when the FO frequency is in the electromechanical range (0.1 – 1 Hz) and the source exhibits high participation in natural modes [5]. Resonance-driven amplification may occur if the source LL is sited close to a generator with high modal participation, and its consumption periodicity matches the frequency of that mode. This work formulates a simulation-based methodology to express the reliability risk from LL-induced FOs in the electromechanical range as a function of FO frequency and load location. The proposed method can be integrated with transmission planning workflows and is summarized in Algorithm 1.

Consider that a planner wishes to assess the impact of load-induced FOs at different frequencies from locations in their territory where interconnection requests are expected. Two types of impacts are to be captured: (a) amplification of power swings within the planner’s footprint, and (b) oscillation spread beyond their area. It is cumbersome to record all output variables in interconnection-scale simulations. Hence, to quantify the impact outside the planning area, oscillation amplitudes for only major tie-lines and generators can be checked. Planners may choose to examine all elements within their area or use a generation capacity or voltage-based screening criteria (e.g., inspect all transmission lines with voltages higher than 230 kV). Model accuracy is critical for accurate risk assessment. Assessment using multiple planning cases representative of different seasonal loading conditions and contingencies can help provide a comprehensive picture of grid vulnerability.

The candidate set of frequencies may comprise ranges specified by data center developers if available, or some frequencies in the electromechanical range can be selected. If poorly damped system modes are observable in the territory,

Algorithm 1 Reliability Impact Assessment Methodology

```

1: Define base power flow and dynamic models
2:  $location\_list \leftarrow$  list of candidate LL locations
3:  $f\_list \leftarrow$  list of candidate oscillation frequencies
4:  $tie\_lines \leftarrow$  list of tie-lines connecting areas
5:  $source\_list \leftarrow$  list of buses and lines to be inspected
   within source area
6:  $A \leftarrow 100$  MW,  $step \leftarrow 20$  MW
7: for  $location$  in  $location\_list$  do
8:   Add a LL model at  $location$  bus
9:   for  $f$  in  $f\_list$  do
10:    Simulate FO with frequency  $f$  and amplitude  $A$ 
11:    Compute amplification factor for all elements in
     $tie\_lines$  and  $source\_list$  using (1)
12:    for elements where amplification > threshold do
13:      Configure protection settings
14:      while Any protection not activated do
15:         $A \leftarrow (A + step)$ 
16:        Rerun simulation with FO amplitude  $A$ 
17:      end while
18:    end for
19:  end for
20: end for

```

then such frequencies should be included. For each candidate location and oscillation frequency, FOs can be simulated per the process described in Section II-B. The next step is to calculate the amplification factor i.e. the maximum power swing observed at a location relative to the source. The amplification factor A_i for element i may be calculated as:

$$A_i = \frac{P_i^{max} - P_i^{min}}{P_{source}^{max} - P_{source}^{min}} \quad (1)$$

As the power system can be linearized around an operating point, the power swing amplitude observed at a location scales linearly with the source oscillation amplitude. As step changes like those observed during AI training cycles may excite multiple system modes, the observed outputs are a superimposition of different frequencies. Computing the amplification factor as described in (1) is simple and avoids the signal processing steps required to isolate specific frequencies from the output.

Amplification factors may be used to rank-order grid elements for further examination. From a wide-area reliability perspective, power swings above 20 MW concern system operators and may be chosen as a threshold. The reliability impact of load-induced FOs may be quantified using several measures, such as, maximum amplification observed inside/outside the source area (a measure of amplification severity), number of grid elements whose amplification factor is above a threshold (a measure of spatial propagation), or a combination thereof.

Planners may also want to determine amplitude limits beyond which protection activation is likely. Configuring relays for all buses and lines in interconnection-scale models is very laborious, and hence we propose setting protection elements only where A_i is higher than a threshold. Modeled protection functions could include under/over frequency, under/over

TABLE II
DOMINANT MODES IN THE 240-BUS WECC MODEL

Frequency (Hz)	0.38	1	0.75	1.3
Damping Ratio (%)	18	7.5	7.5	5.3

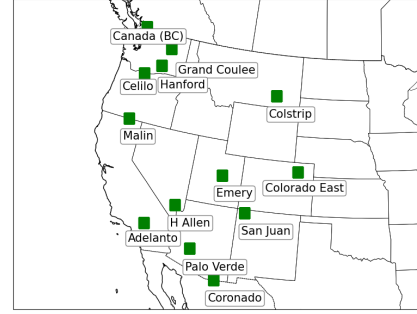


Fig. 2. Large load locations modeled in the 240-bus test system.

voltage, and zone 3 distance protection. Existing remedial action schemes and out-of-step relays for generators can also be represented. Once the protection elements are configured, oscillations with increasing amplitudes may be simulated until at least one protection is engaged, yielding an upper limit on permissible oscillation amplitudes.

IV. RESULTS

This section describes findings from applying the proposed risk assessment framework to the 240-bus WECC model [7]. All simulations were performed using PSS/E, and the workflow was automated using the Python API.

A. Test System Description

The 240-bus reduced WECC model comprises about 140 GW of generation with a realistic fuel-mix. Due to the aggregation of generators, detailed generator interactions cannot be observed in this model. Additionally, inferences from this model may not directly translate to the full WECC system. Nevertheless, because the model is publicly available, the dissemination of observations and their implications is easy.

As the oscillation excitation risk is correlated with the properties of the system's natural modes, the modes of the 240-bus system were estimated using the procedure outlined in [17]. The estimated frequencies and damping ratios of the observed dominant modes are listed in Table II.

B. Observations

Thirteen locations across the system were selected as potential LL sites, as shown in Fig. 2. LLs were represented by the ZIP model. Three types of load perturbations were simulated for each site - (i) 0.5 Hz square wave, (ii) 1 Hz square wave, and (iii) bi-periodic load pattern with the slow and fast frequencies as 0.1 and 1 Hz, respectively. Maximum FO amplification observed for each site is summarized in Fig. 3. Fig. 4 offers a closer look at the impact of FOs injected from the H ALLEN location in Nevada - only elements where $A_i > 0.2$ are plotted. It can be seen that the FO spatial spread is wider at lower frequencies.

Fig. 3 offers several insights. First, oscillation amplification, and hence the reliability risk is higher for 1-Hz oscillations

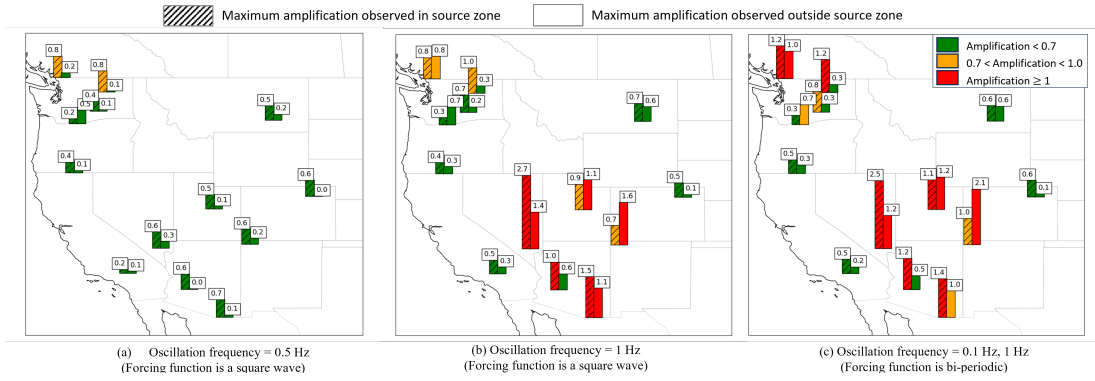


Fig. 3. Maximum amplification observed inside and outside source zone when FOs are injected from LLs at different WECC locations

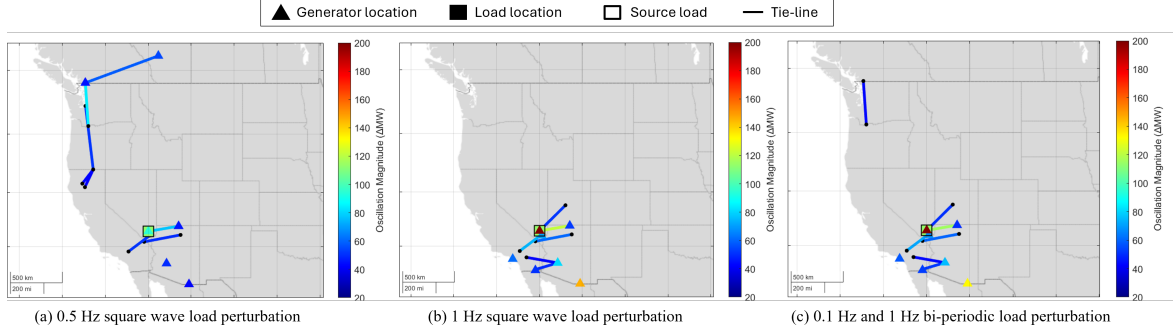


Fig. 4. Grid elements where oscillations higher than 20 MW are recorded when 100 MW load perturbations are introduced from the H ALLEN bus

than 0.5-Hz ones. This is expected as a dominant mode exists at 1 Hz. Further, the 1-Hz FO has the highest amplification when injected from locations in the Southwest. This suggests that if LLs in this region were to oscillate at 1 Hz, they could pose a significant risk to grid reliability, necessitating mitigation measures. Fig. 3 also shows that the oscillation amplification risk increases in the (0.1,1) Hz bi-periodic perturbation case, possibly because additional locations in the hydro-rich Pacific Northwest show high excitability at 0.1 Hz.

1) *1 Hz Mode*: Fig. 3 shows that resonance-driven amplification occurs when 1-Hz oscillations are simulated from LLs in the Southwest. To explain this phenomenon, the corresponding mode shape plot may be inspected (Fig. 5). As seen from the figure, the 1-Hz mode is observable mainly in the Southwest, with generators in Nevada swinging against those in Southern California, Arizona, and New Mexico. The mode shape magnitude, a measure of mode observability, is highest in the H ALLEN bus, and amplification is highest when a FO is injected from this location. The power swings at different generators in response to a 1-Hz perturbation from H ALLEN is presented in Fig. 6. A linear relationship is observed between the FO source's mode shape magnitude and observed maximum amplification for oscillations injected at modal frequency, as shown in Fig. 7. This is a useful insight, because transmission planners can use mode shape magnitudes as an initial screening criterion for identifying risky load cycling frequencies when data center interconnection requests are received at a location.

2) *FO Frequencies Near System Modes*: The 0.38-Hz mode in the 240-bus model, representative of the WECC North-

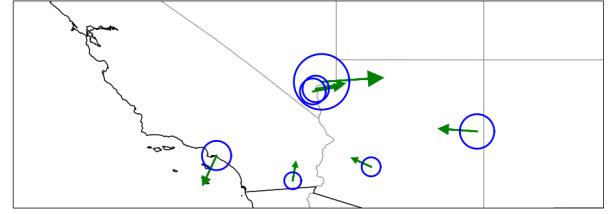


Fig. 5. 1-Hz mode shape plot. Circles mark generator locations. Radii are proportional to mode shape magnitudes (measure of observability). Arrows show mode shape angles. Generators with similar angles swing together.

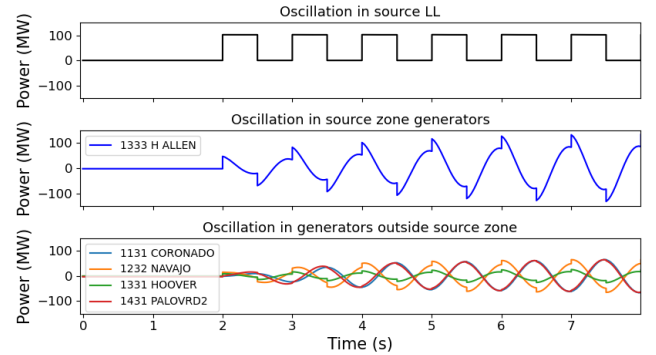


Fig. 6. Oscillations observed in generator power outputs when 1 Hz, 100 MW load perturbations are introduced from the H ALLEN bus. Mean values subtracted for easy visualization.

South (N-S) mode, is the most geographically widespread. Hence, to simulate a particularly concerning resonance case, a 100-MW load perturbation matching this mode frequency was introduced from a LL placed at the CANADA bus in the British Columbia (BC) zone. The CANADA bus was observed to have a high mode shape magnitude for the N-S mode. Sys-

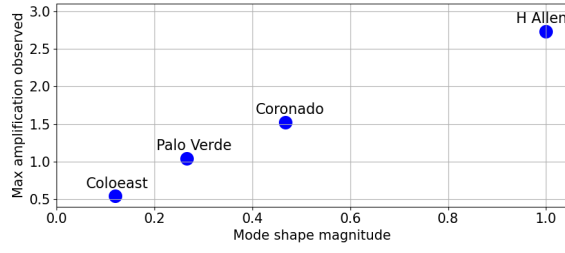


Fig. 7. Max. oscillation amplification vs. mode shape magnitude of source

TABLE III
OSCILLATION VIOLATIONS (> 20 MW) FOR 100 MW LOAD PERTUB.

Category	H ALLEN at 1 Hz		CANADA at 0.38 Hz	
	P-Gen	P-Flow	P-Gen	P-Flow
# Inst. >20 MW	11	8	8	13
# Inst Outside Source Zone	10	5	6	11
Max. p-p Amplitude in Source Zone (MW)	280.93	122.83	149.06	161.58
Max. p-p Amplitude Outside Source Zone (MW)	139.57	58.45	98.25	75.48
Max. Amplitude Locn. in Source Zone	H ALLEN	MEAD – H ALLEN	CANAD G1	NORTH – CANADA
Max. Amplitude Locn. Outside Source Zone	CORONADO	MEAD – VICTORVYL	CMAIN GM	MALIN – MERIDIAN

tem elements where appreciable power swings are observed for this scenario are plotted in Fig. 8. Large oscillations higher than 100 MW peak-to-peak are observed on key transmission lines in the Northwest corridor connecting Canada, the Pacific Northwest, and Northern California. Moreover, power swings greater than 20 MW are observed in generators as far south as Arizona, even though the N-S mode is well-damped.

Table III lists where high amplification is observed when 1-Hz and 0.38-Hz FOs are introduced from LLs in the H ALLEN and CANADA buses, respectively. The highest oscillation amplification observed for the 0.38 Hz case is lower, perhaps attributable to the very high damping factor in the examined case. Despite this, large power swings are observed in fewer elements for the 1-Hz case compared to the 0.38-Hz case, due to the wider geographical observability of the N-S mode.

V. CONCLUSION

This paper presents a simulation-based methodology to automate the assessment of risks from LL-induced forced oscillations. The framework streamlines the process for screening potential LL interconnection points to identify high-risk locations where FO–natural mode resonance and the wide-area propagation/amplification of induced oscillations are probable. The proposed approach is used to evaluate the reliability impact of LL-induced oscillations over a range of frequencies from 13 points in the 240-bus WECC model. Insights from this study can help transmission planners initiate detailed studies or recommend oscillation mitigation measures for high-risk interconnection locations. The LL operators can also be educated on the wide-area reliability risks posed by load variations at certain frequencies, prompting software-side mitigation measures where possible.

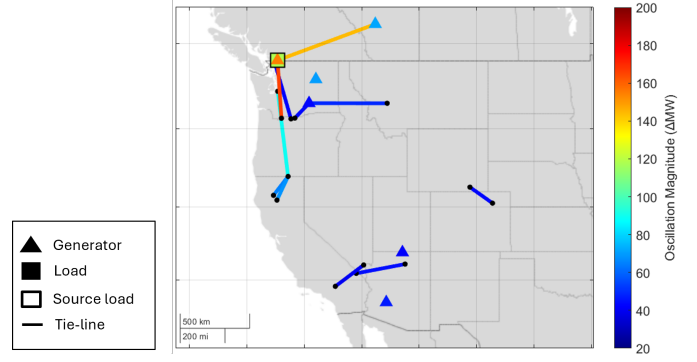


Fig. 8. Grid elements where > 20 MW oscillations are observed when 0.38 Hz, 100 MW load perturbations are introduced from the CANADA bus

The paper also outlines how to select a subset of grid elements whose protection settings must be configured to determine oscillation amplitudes that do not trigger relays. This greatly simplifies workflow by eliminating the need to configure and debug thousands of elements in interconnection-scale models while still capturing the impacts of LL-induced power swings. In future work, the authors aim to investigate the potential of low-frequency oscillation-induced relay misoperations with realistic interconnection-scale models.

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